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(54) **ASYMMETRICAL HEEL PAD**

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(51) **Int. Cl.**

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(2013.01); **A43B 7/144** (2013.01); **A43B 7/149**
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(58) **Field of Classification Search**

CPC **A43B 23/08**; **A43B 7/14**; **A43B 7/1405**;
A43B 23/088; **A43B 7/141**; **A43B 7/1464**

USPC 36/69
See application file for complete search history.

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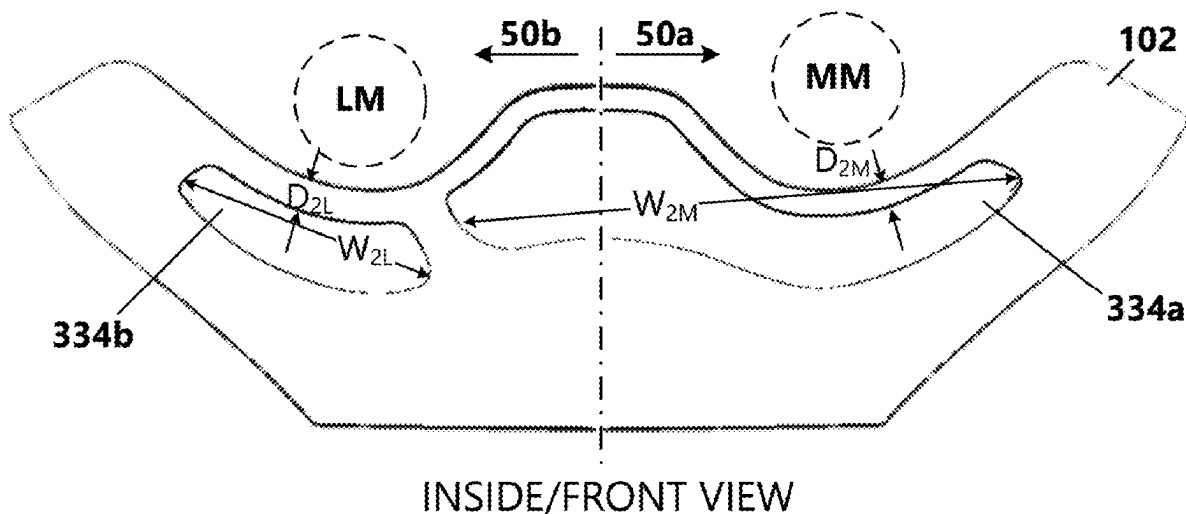
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(57) **ABSTRACT**

Systems and methods for providing a golf shoe comprising a heel cup with an asymmetrical padding configuration to reduce foot movement during a golf swing. The asymmetrical padding configuration may include one or more heel pads that fit the contour of the heel area of the wearer's foot to provide proper support and fit to a golfer to reduce the voids of space that normally occur around the heel portion in a shoe. The support may lead to a better fit for increased comfort and more stability during the golf swing by reducing foot movement during the swing. The heel pad(s) may be provided between an outer and inner layer of material forming the shoe's upper and may be made of a variety of different materials that may give different stiffness and comfort to the heel area of the wearer's foot. In some examples, the shoe may be waterproof.

20 Claims, 12 Drawing Sheets



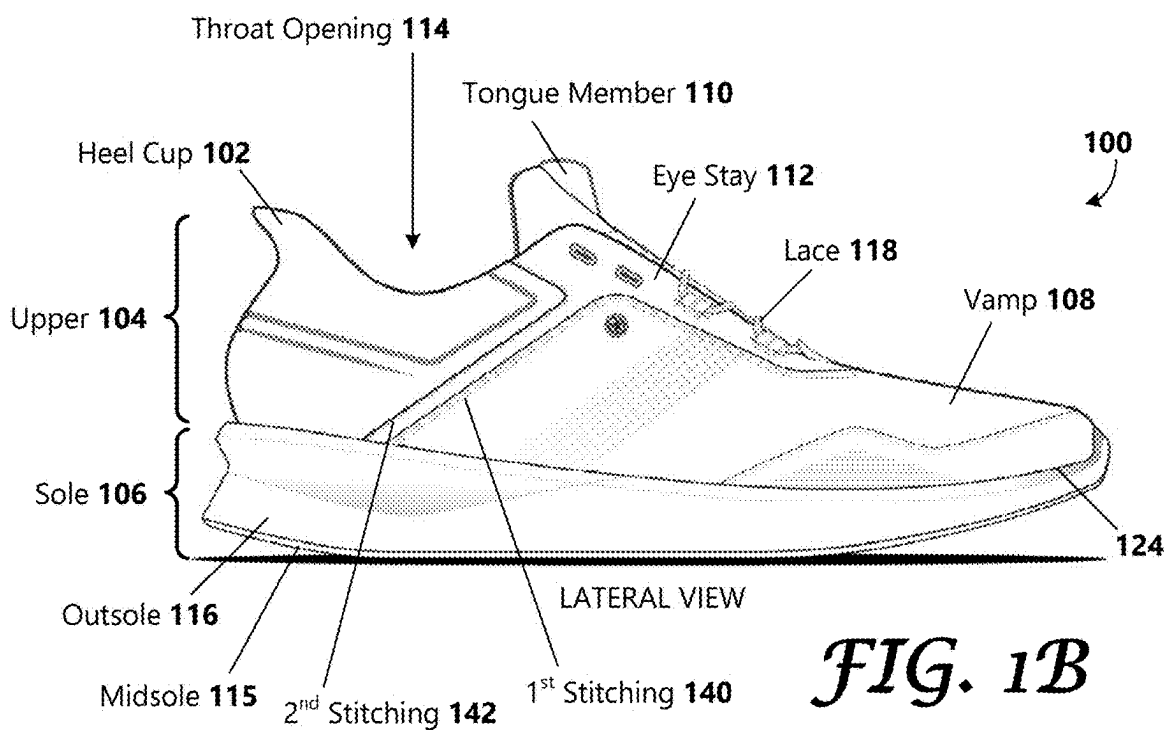
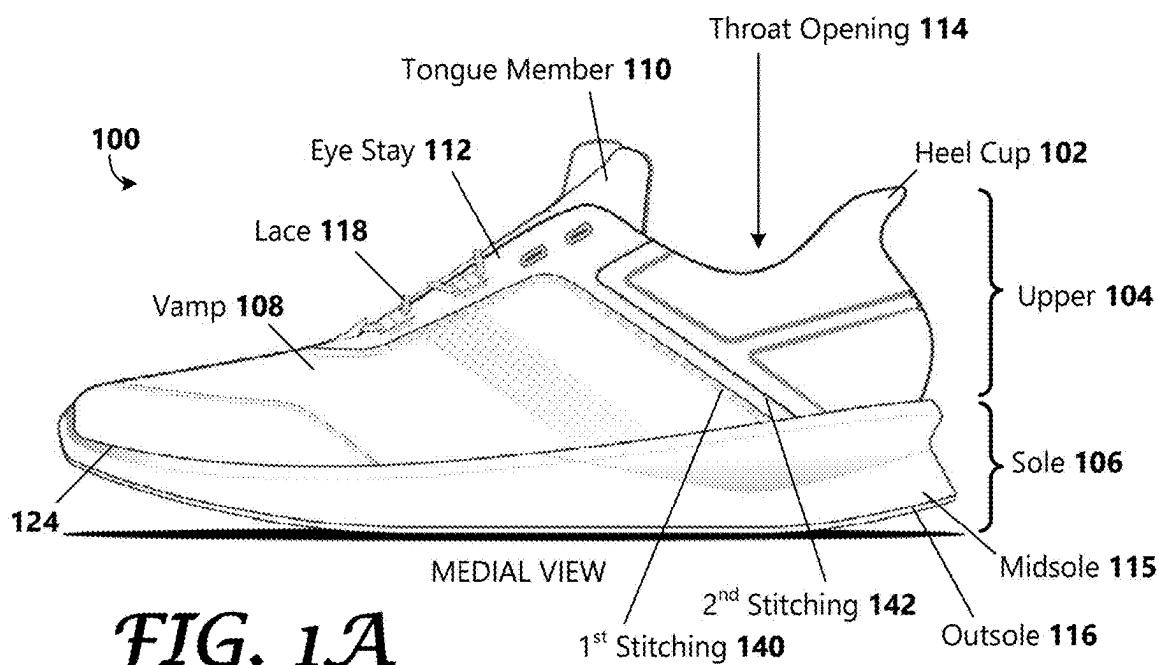
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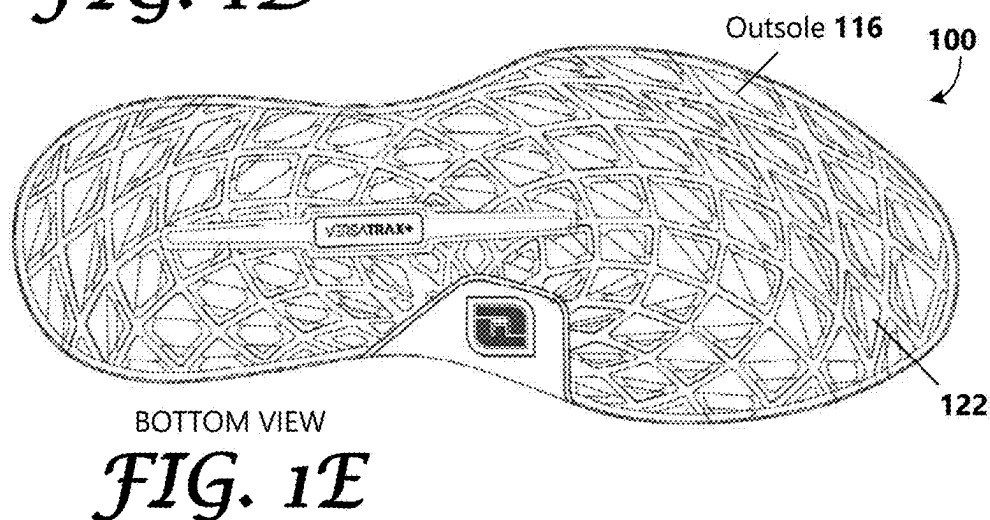
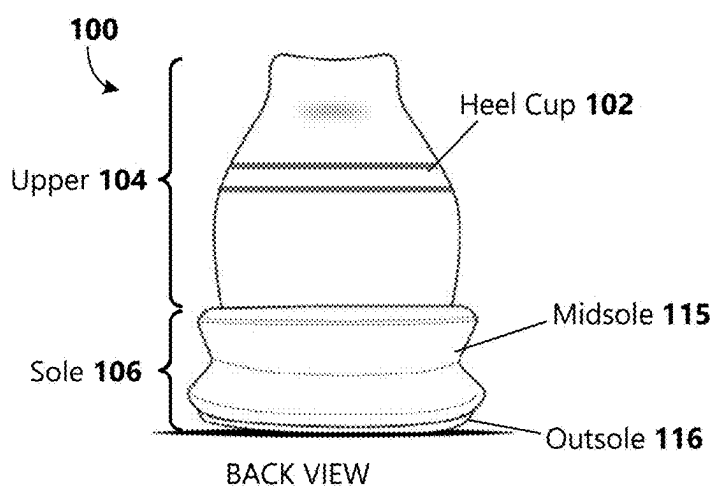
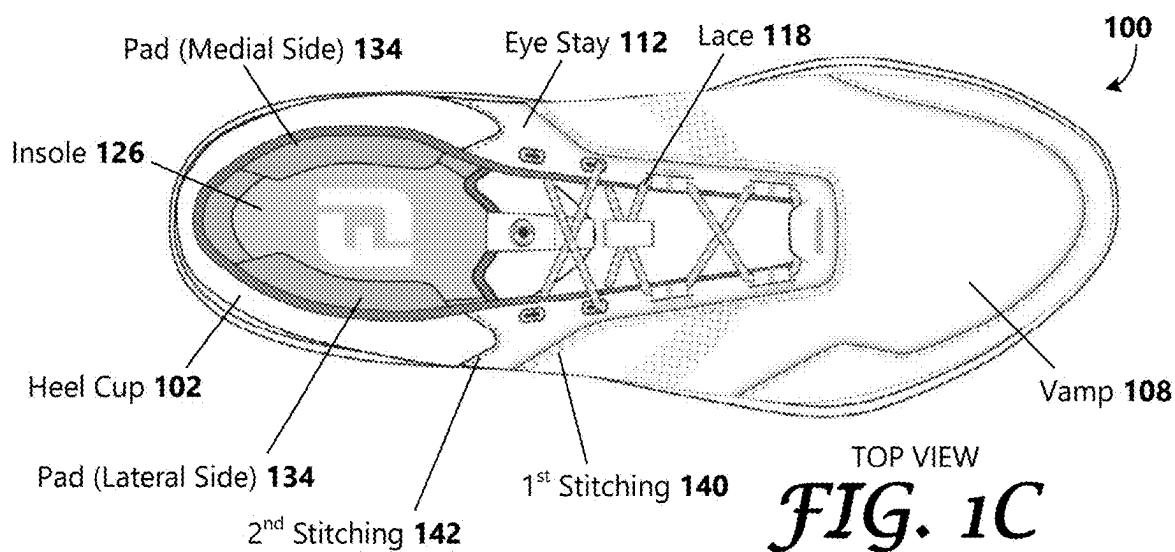
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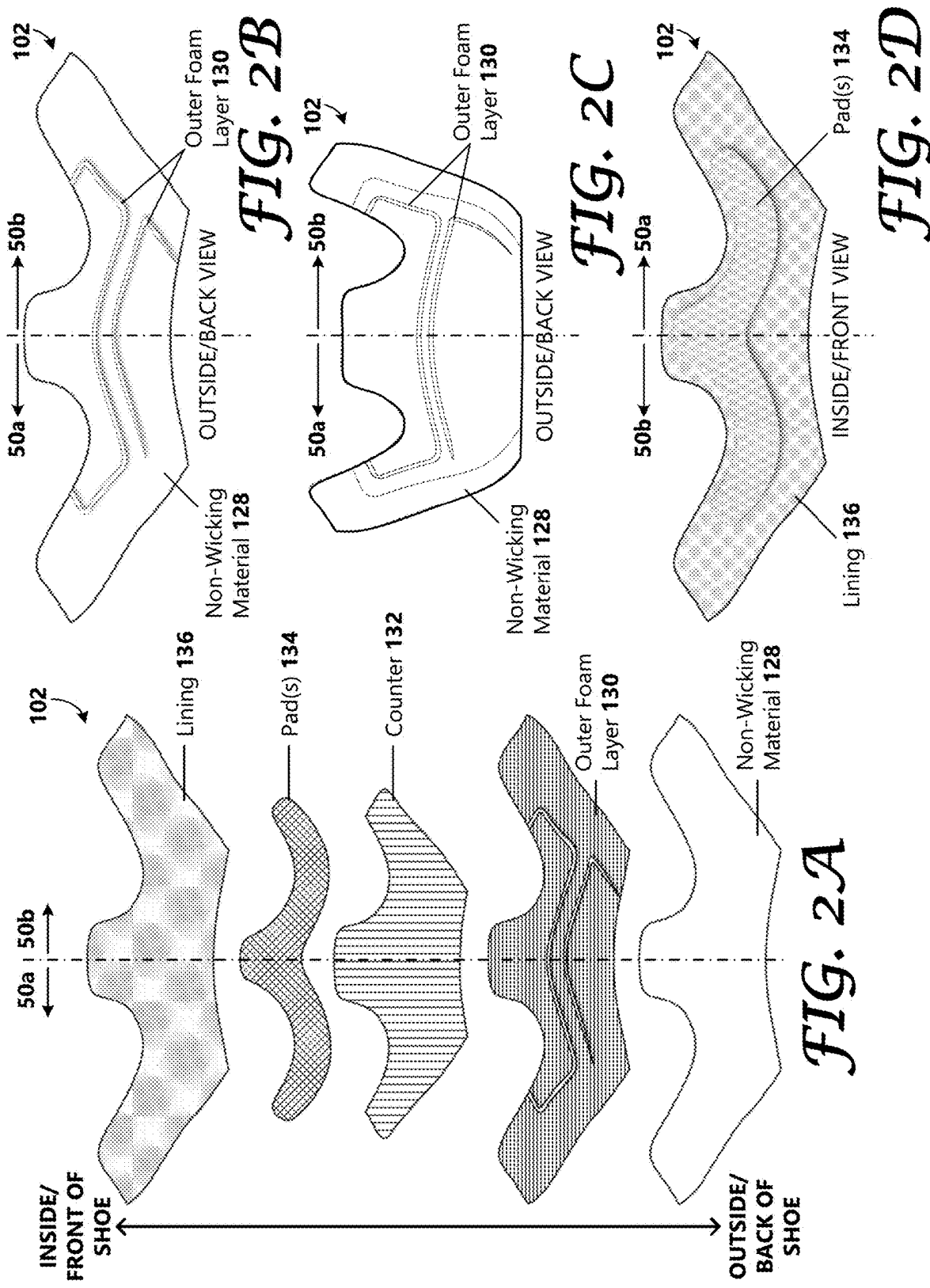
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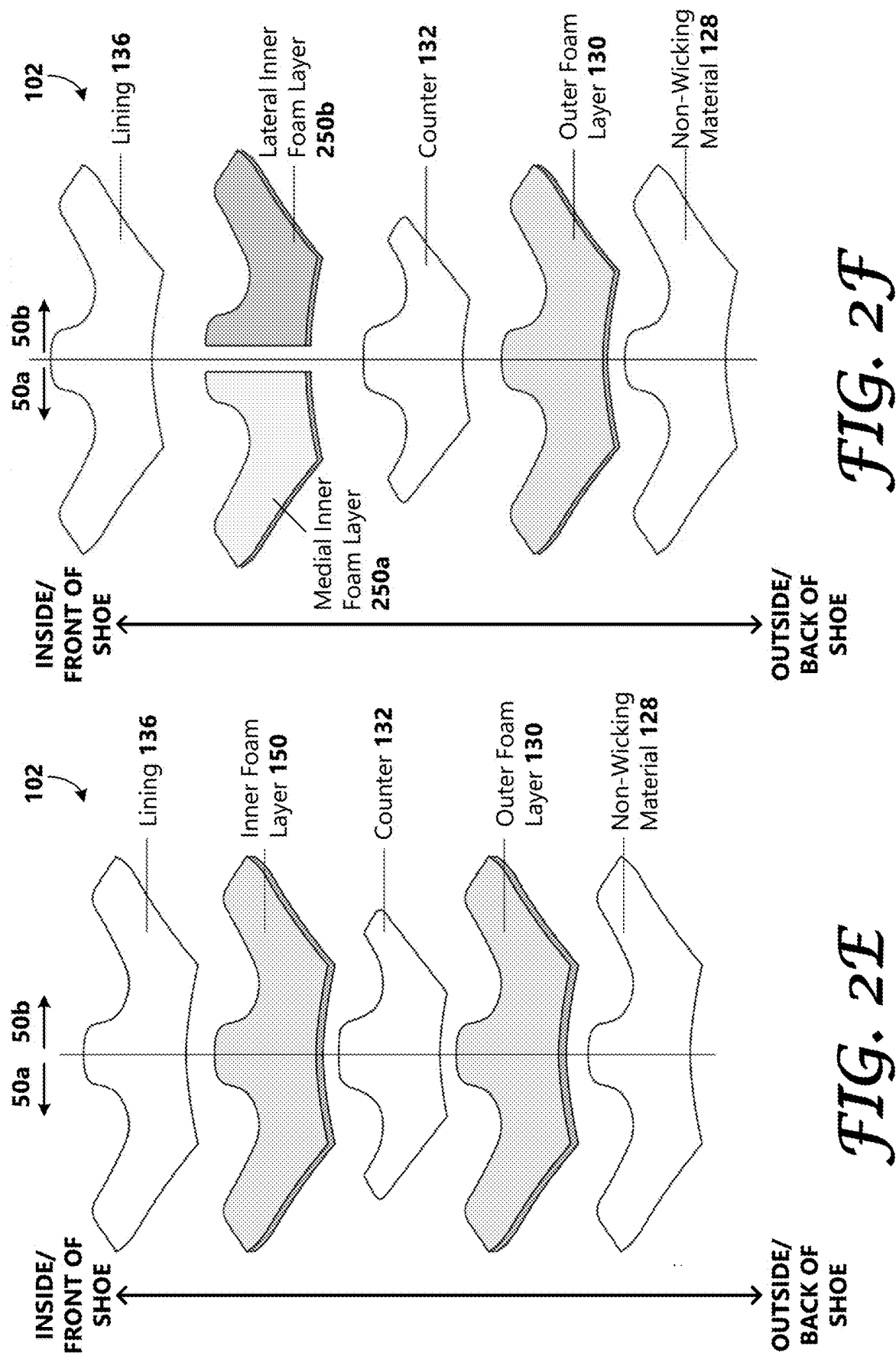
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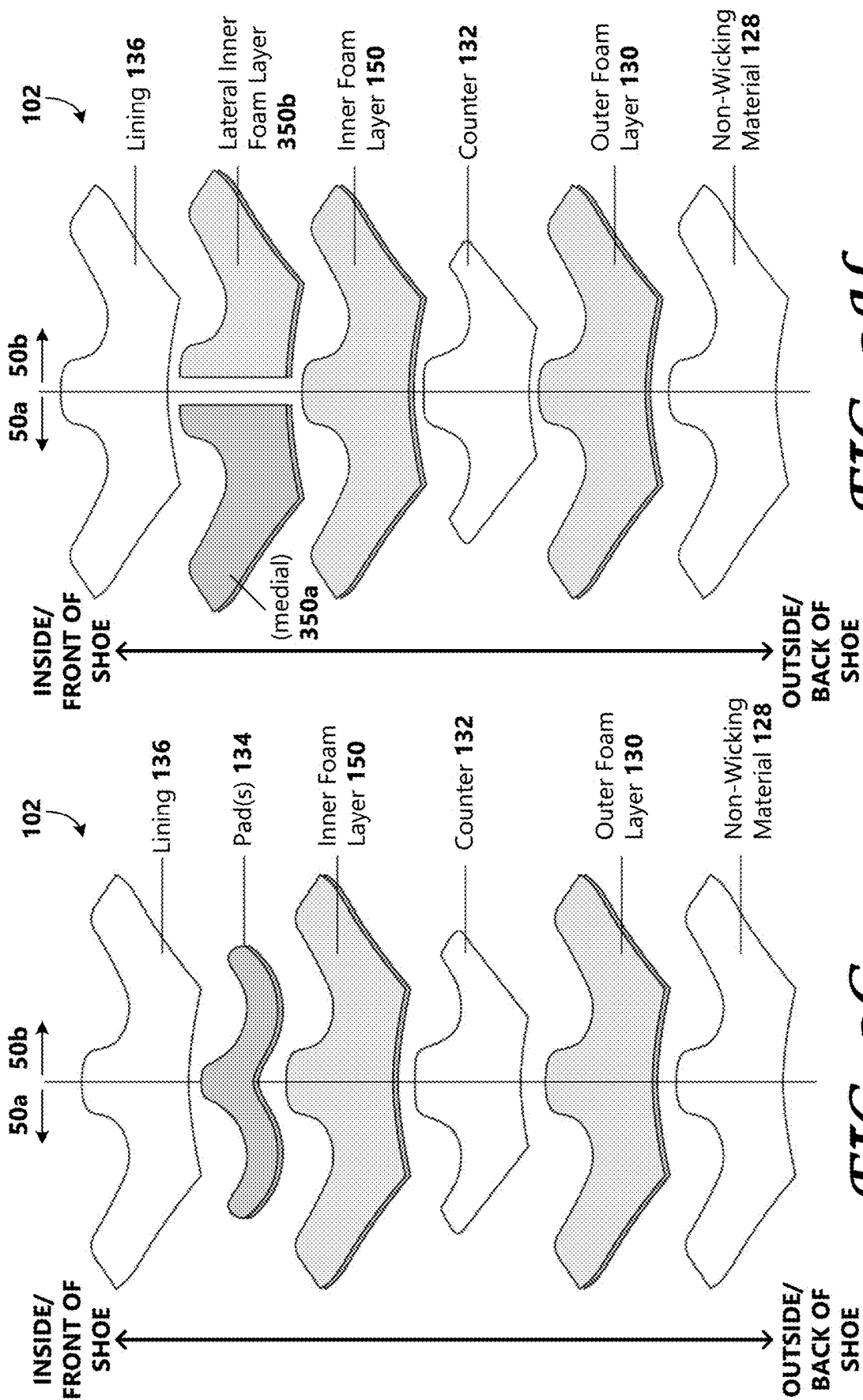


FIG. 2H

FIG. 2G

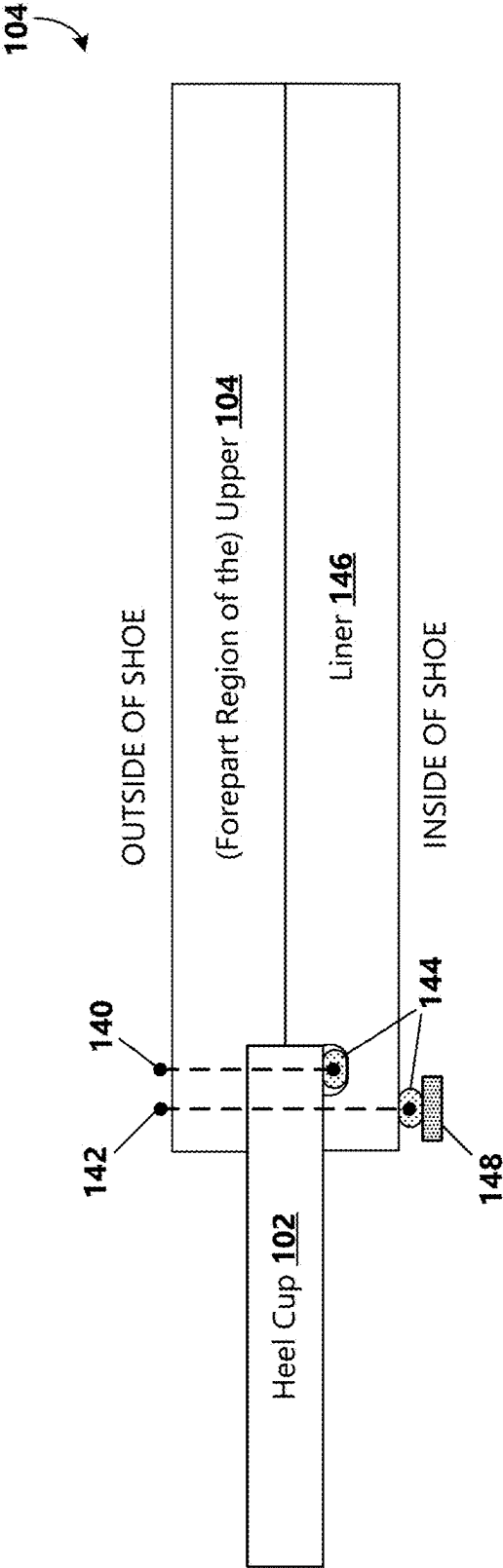
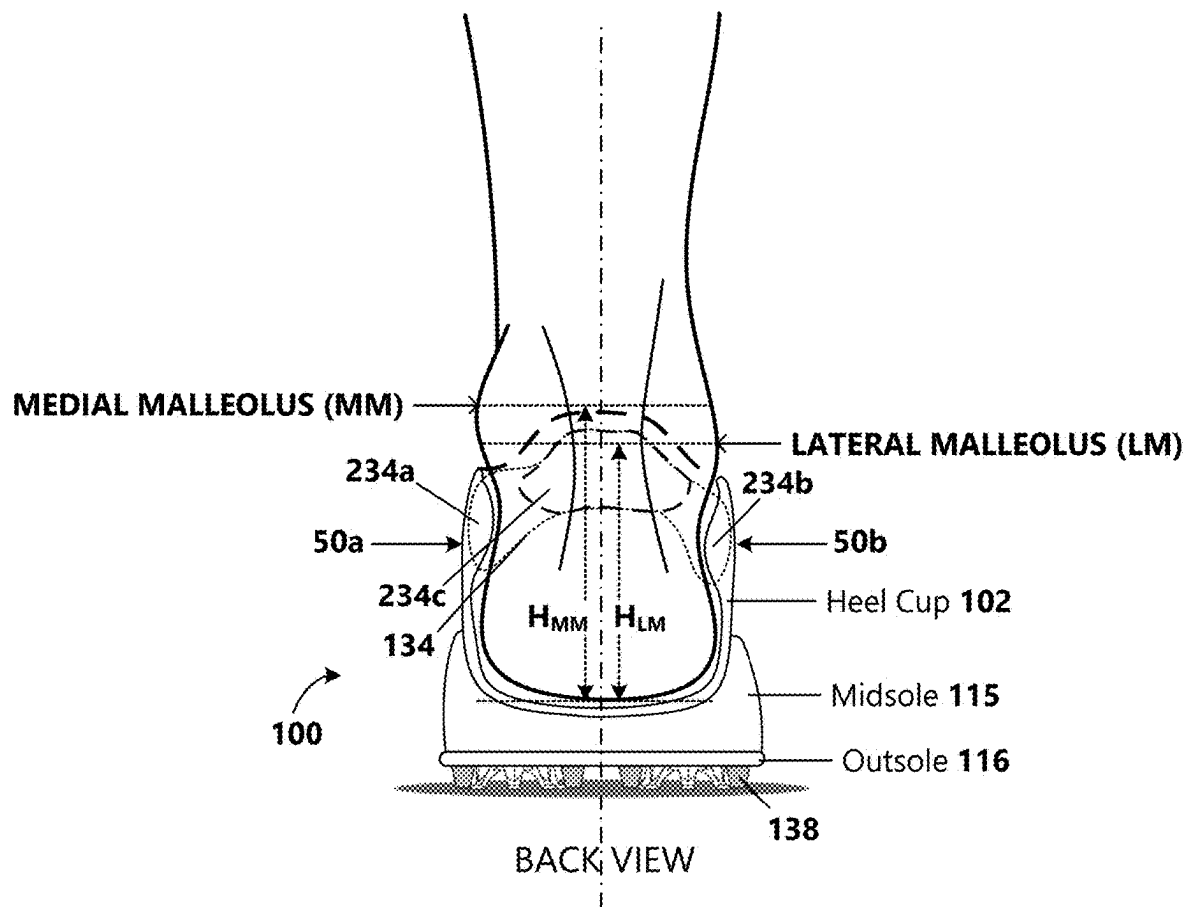
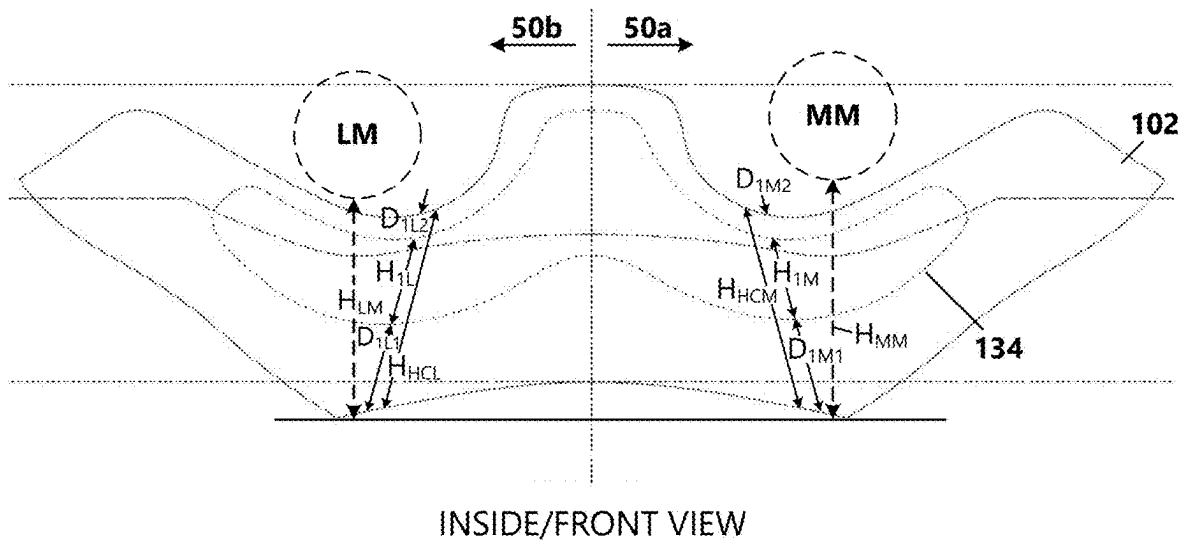
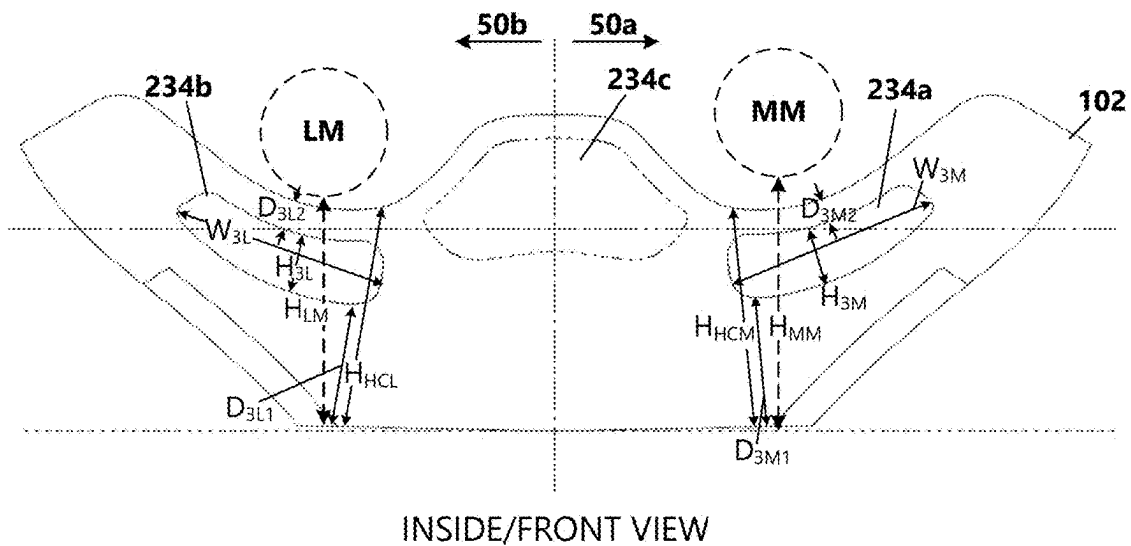


FIG. 3

*FIG. 4*

*FIG. 5A**FIG. 5B*

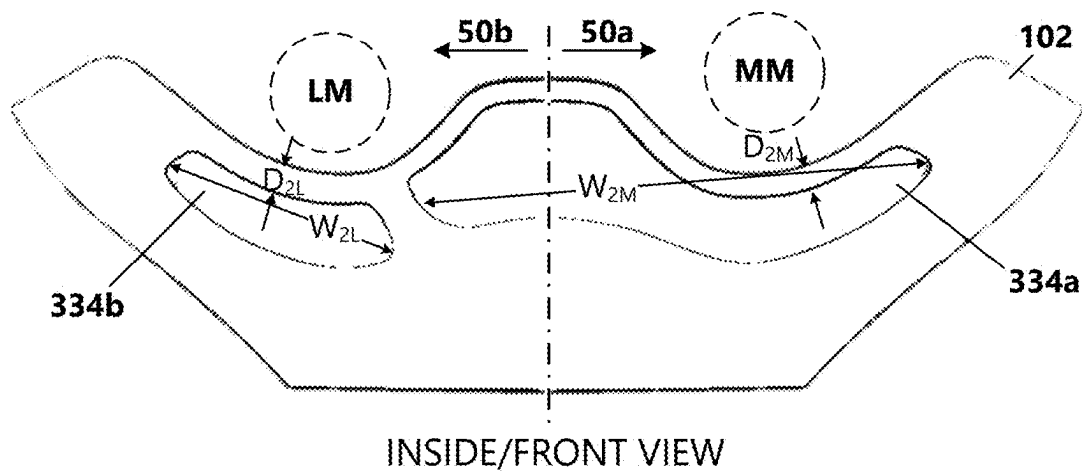


FIG. 5C

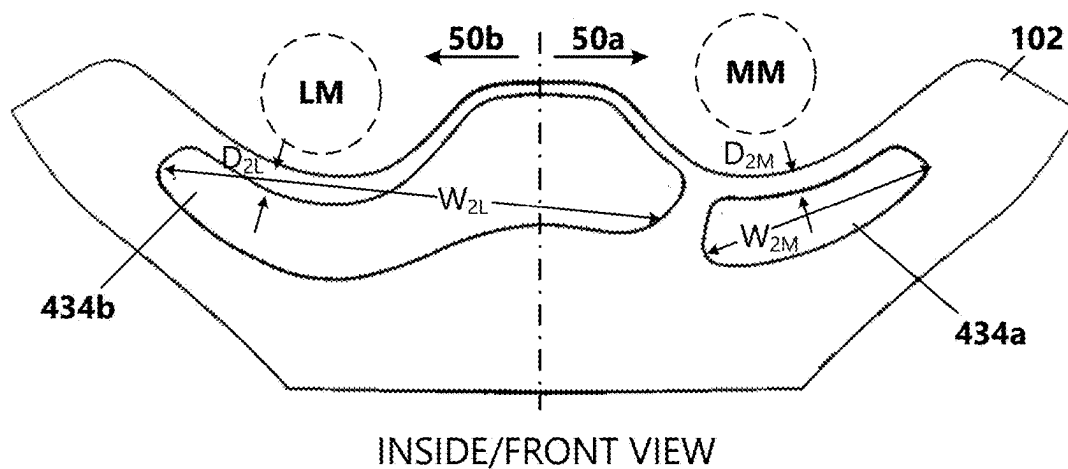
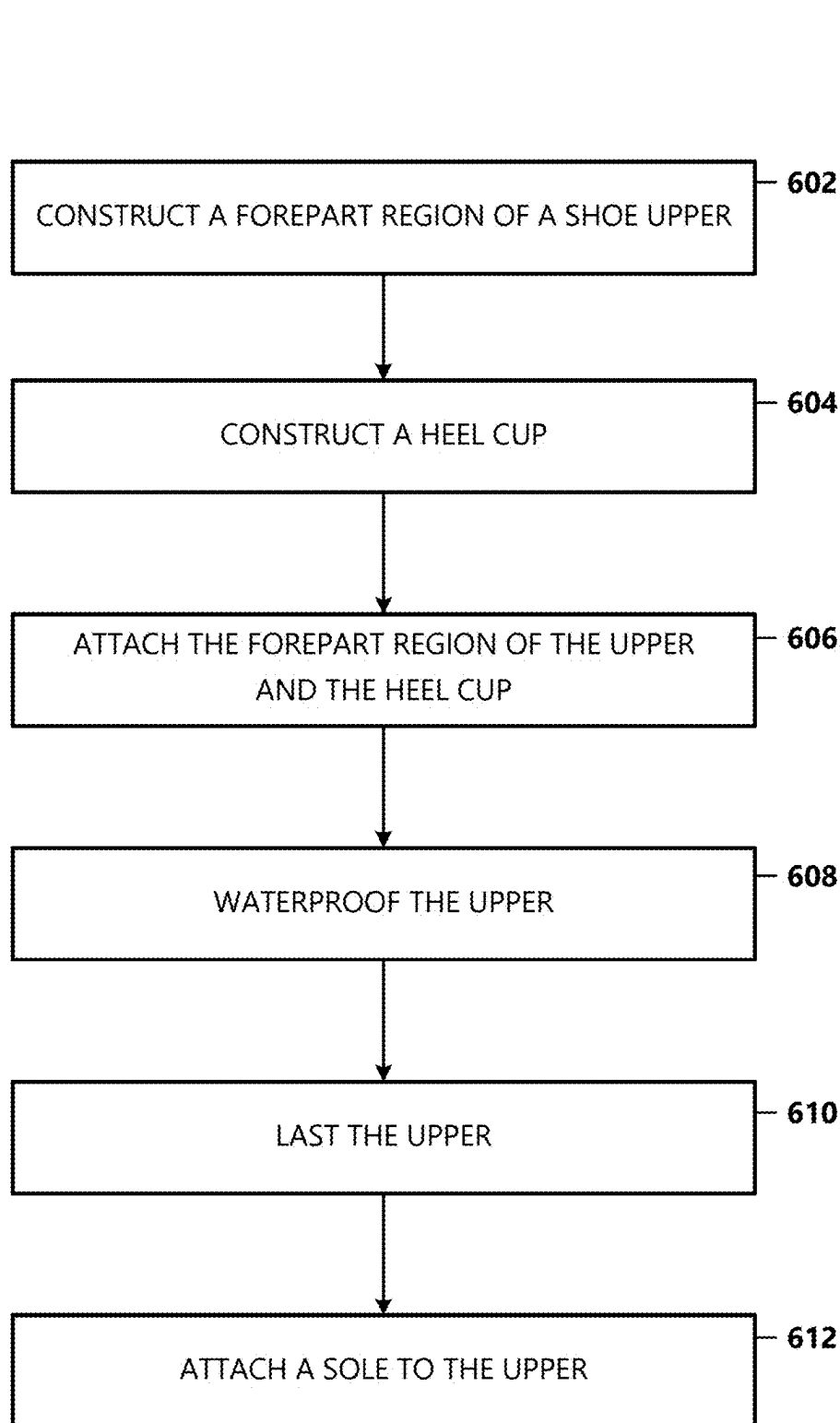
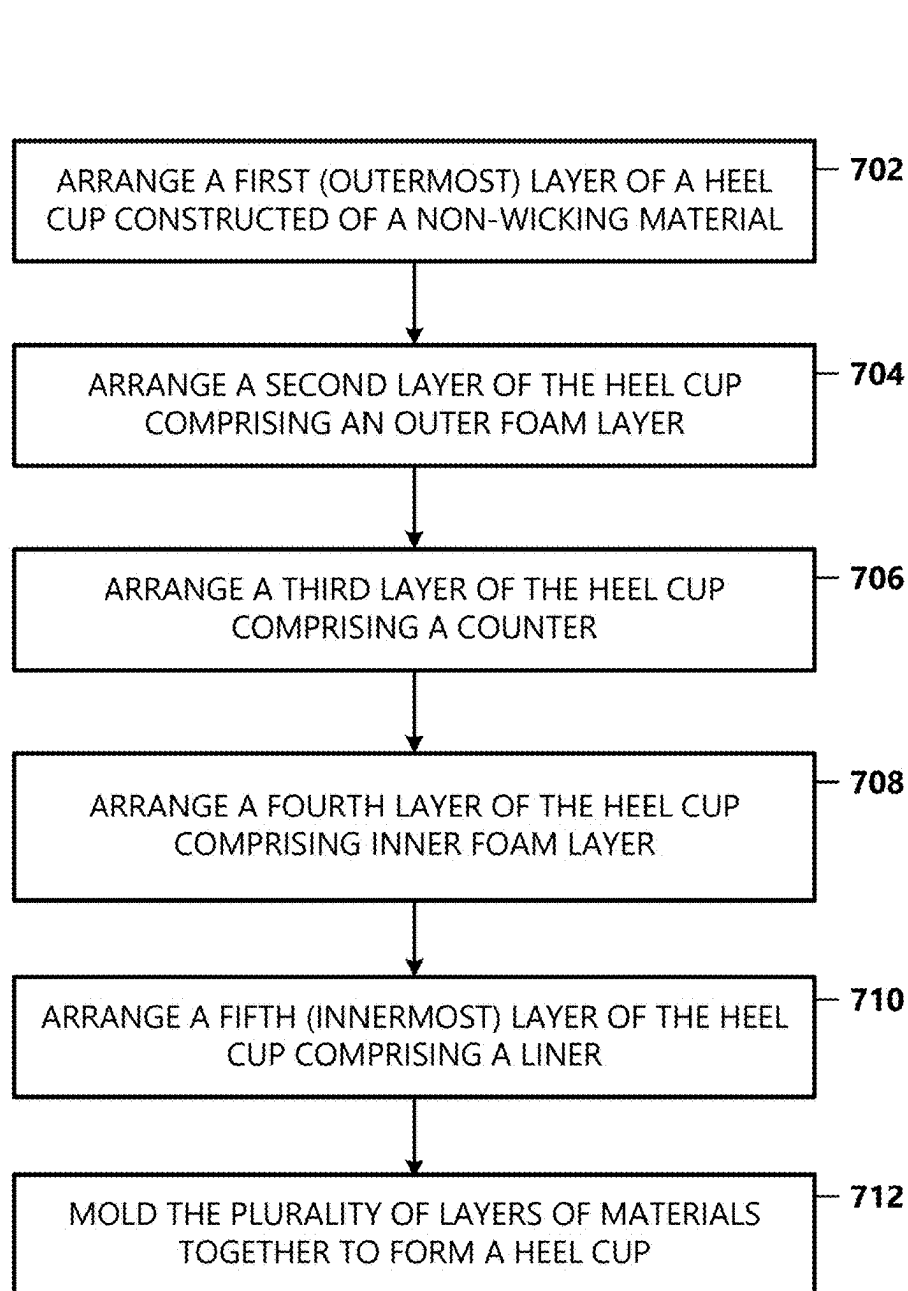
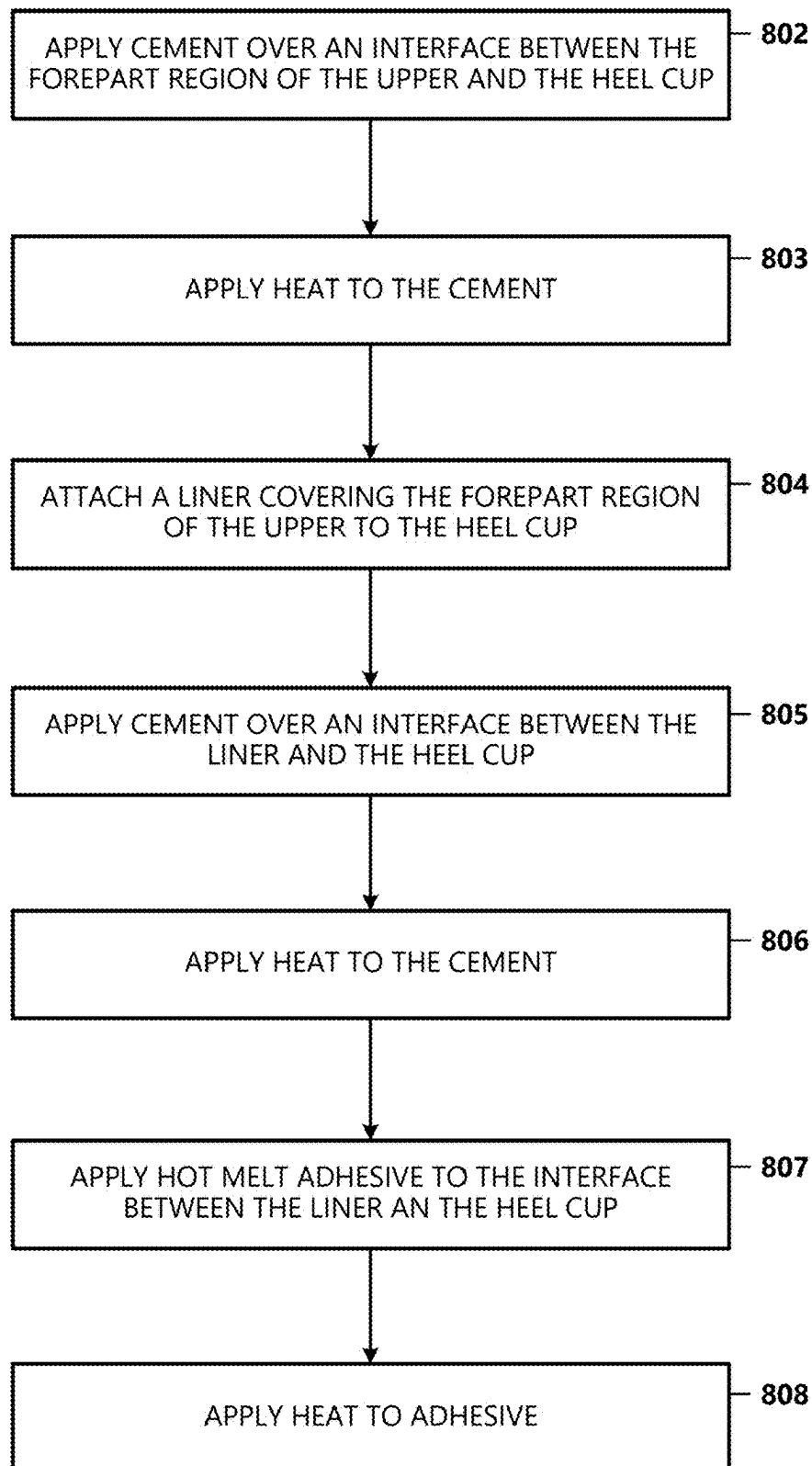


FIG. 5D

*FIG. 6*

*FIG. 7*

*FIG. 8*

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ASYMMETRICAL HEEL PAD**CROSS-REFERENCE TO RELATED APPLICATION**

This application is a continuation of U.S. patent application Ser. No. 17/529,864, filed on Nov. 18, 2021, which is incorporated herein by reference. This application claims priority to the above-referenced application.

BACKGROUND

Having the proper equipment when playing a sport can be a factor in how well an athlete may perform. For example, proper equipment in the sport of golf may allow a golfer to hit the golf ball longer, straighter, and more consistently, thus improving the golfer's performance and overall score. Golf shoes are one example piece of equipment that can substantially affect a golfer's performance. For example, traction, stability, and support provided by a golfer's shoes can affect how the golfer is able to perform the body movements necessary to culminate in an ideal contact between a club head and a ball (i.e., a golf swing). If the shoes do not fit the golfer properly, the golfer's foot can move relative to the ground, which is undesirable. For example, increased movement of the foot during the golf swing can lead to less power during the swing and less accuracy in the golf shot.

It is with respect to these and other general considerations that the aspects disclosed herein have been made. Also, although relatively specific problems may be discussed, it should be understood that the examples should not be limited to solving the specific problems identified in the background or elsewhere in this disclosure.

SUMMARY

Examples of the present disclosure describe a golf shoe comprising a heel cup comprising an asymmetrical padding configuration to reduce foot movement during a golf swing. The asymmetrical padding configuration may include one or a plurality of heel pads that fit the contour of the heel area of the wearer's foot. For example, the one or more heel pads may have an asymmetrical configuration so that the heel area provides proper support and fit to a golfer to reduce the voids of space that normally occur around the heel portion in a shoe. The support may lead to a better fit for increased comfort and more stability during the golf swing by reducing foot movement during the swing. The decreased movement of the foot during the golf swing can lead to more power during the swing and greater accuracy in the golf shot.

In an example, the technology relates to a golf shoe comprising: an upper and a sole attached to the upper, the upper comprising: a heel cup with a first surface and a second surface; and a plurality of asymmetrically shaped or positioned pads provided between the first surface and the second surface, the plurality of asymmetrically shaped or positioned pads comprising a first pad and a second pad that is entirely separated from the first pad, wherein the first pad is configured to extend along a first side of the heel cup and across a center line onto a second side of the heel cup, and wherein the second pad is positioned entirely on the second side of the heel cup between the center line of the heel cup and a front or side edge of the second side of the heel cup.

This Summary is provided to introduce a selection of concepts in a simplified form that are further described

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below in the Detailed Description. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Additional aspects, features, and/or advantages of examples will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

Non-limiting and non-exhaustive examples are described with reference to the following figures.

FIG. 1A depicts a medial side view of an example golf shoe in which an asymmetrical heel pad configuration may be implemented.

FIG. 1B depicts a lateral side view of the example golf shoe of FIG. 1A.

FIG. 1C depicts a top view of the example golf shoe of FIG. 1A.

FIG. 1D depicts a back view of the example golf shoe of FIG. 1A.

FIG. 1E depicts a bottom view of the example golf shoe of FIG. 1A.

FIG. 2A depicts an exploded view of various layers that may be used to form an example heel cup for the example golf shoe of FIG. 1A according to an example.

FIG. 2B depicts an outside (or back) view of the various layers of the example heel cup of FIG. 2A arranged together in a flattened form.

FIG. 2C depicts an outside (or back) view of the example heel cup of FIG. 2A arranged together in a curved form.

FIG. 2D depicts an inside (or front) view of the various layers of the example heel cup of FIG. 2A arranged together in a flattened form.

FIG. 2E depicts an exploded view of various example layers that may be used to form an example heel cup for the example golf shoe of FIG. 1A according to another example.

FIG. 2F depicts an exploded view of various example layers that may be used to form an example heel cup comprising a dual density asymmetrical heel pad configuration for the example golf shoe of FIG. 1A according to another example.

FIG. 2G depicts an exploded view of various example layers that may be used to form an example heel cup comprising a pre-formed asymmetrical heel pad configuration for the example golf shoe of FIG. 1A according to another example.

FIG. 2H depicts an exploded view of various example layers that may be used to form an example heel cup comprising a triple density asymmetrical heel pad configuration for the example golf shoe of FIG. 1A according to another example.

FIG. 3 depicts a schematic diagram of an upper of the example golf shoe including waterproofing features.

FIG. 4 depicts a back view of an example golf shoe shown on a wearer's foot.

FIG. 5A depicts an inside (or front) view of an example single heel pad configuration.

FIG. 5B depicts an inside (or front) view of an example multiple heel pad configuration including three (3) heel pads.

FIG. 5C depicts an inside (or front) view of an example multiple heel pad configuration including two (2) heel pads.

FIG. 5D depicts an inside (or front) view of another example multiple heel pad configuration including two (2) heel pads.

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FIG. 6 depicts example operations of a method of making a golf shoe comprising an asymmetrical heel pad configuration according to an embodiment.

FIG. 7 depicts example operations of a method of constructing an asymmetrical configured heel pad according to an embodiment.

FIG. 8 depicts example operations of a method of waterproofing a shoe upper according to an embodiment.

DETAILED DESCRIPTION

The present invention now will be described more fully in reference to the accompanying figures, in which embodiments of the invention are shown. However, this invention should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the invention to those skilled in the art. In the drawings, like numbers refer to like elements throughout. Thicknesses and dimensions of some components may be exaggerated for clarity. The views shown in the Figures are of a right shoe and it is understood the components for a left shoe will be mirror images of the right shoe. It also should be understood that the shoe may be made in various sizes and thus the size of the components of the shoe may be adjusted depending upon the shoe size.

The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the invention. As used herein, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

It will be understood that when an element is referred to as being “attached,” “coupled” or “connected” to another element, it can be directly attached, coupled or connected to the other element or intervening elements may also be present. In contrast, when an element is referred to as being “directly attached,” “directly coupled” or “directly connected” to another element, there are no intervening elements present.

It is noted that any one or more aspects or features described with respect to one embodiment may be incorporated in a different embodiment although not specifically described relative thereto. That is, all embodiments and/or features of any embodiment can be combined in any way and/or combination. Applicant reserves the right to change any originally filed claim or file any new claim accordingly, including the right to be able to amend any originally filed claim to depend from and/or incorporate any feature of any other claim although not originally claimed in that manner. These and other objects and/or aspects of the present invention are explained in detail in the specification set forth below.

As discussed above, golf shoes are an important part of a player's equipment, and lack of stability in the golf shoe may negatively impact a player's performance. One way in which golf shoes may not fit the golfer properly is in the heel area, where voids of space may be formed below the bony prominences located on each side of the golfer's ankles. Such voids of space occur commonly around the heel in golf shoes due at least in part to the asymmetrical configuration of the bony prominences between each side of the human ankle and the typically symmetrical configuration of the heel area of golf shoes. For example, each leg of the human body is supported by two bones: a tibia on the medial or inner side of the leg and the fibula on the lateral or outer side of the leg, and the bony prominence on each side of the human ankle is referred to as a malleolus. A medial malleolus is formed

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by the lower end of the tibia and is the prominence on the medial side of the ankle. A lateral malleolus, the prominence on the lateral side of the ankle, is formed by the lower end of the fibula and descends lower than the medial malleolus.

Currently, padding provided in the heel area of golf shoes is typically symmetrical in design and does not account for the asymmetrical configuration of the medial and lateral malleolus. Thus, the symmetrical design of the padding may form a void in the heel area around the medial and/or lateral malleolus that may allow for movement of the wearer's foot. It may be desirable, therefore, to provide a golf shoe that fits the contour of the wearer's heel area for reducing foot movement during a golf swing.

Additionally, given that the sport of golf is an outdoor sport, a golfer may sometimes play in wet and/or soggy conditions. The sport of golf includes stretches of walking, which if a golfer's shoes are not waterproof, the golfer's foot can get wet. As can be appreciated, this may not only be uncomfortable to the golfer, but in certain locations and/or during low temperature conditions, can be unsafe for the golfer. It may be desirable, therefore, to provide a golf shoe that is waterproof for keeping the wearer's foot dry during wet or soggy conditions.

To help alleviate the above problems, the examples of the present disclosure describe a golf shoe comprising an asymmetrical heel padding configuration to reduce foot movement during a golf swing. The asymmetrical heel padding configuration may include one or a plurality of heel pads that fit the contour of the heel area of the wearer's foot. For example, the one or more heel pads may have an asymmetrical configuration so that the heel area provides proper support and fit to a golfer to reduce the voids of space that normally occur around the heel portion in a shoe. The support may lead to a better fit for increased comfort and more stability during the golf swing by reducing foot movement during the swing. The decreased movement of the foot during the golf swing can lead to more power during the swing and greater accuracy in the golf shot.

FIGS. 1A-1E depict various views of an example golf shoe 100, sometimes referred to herein generally as a shoe, in which aspects of a heel cup 102 comprising an asymmetrical heel padding configuration may be implemented. For example, FIG. 1A is a medial (e.g., inner) side view of the shoe 100, FIG. 1B is a lateral (e.g., outer) side view of the shoe 100, FIG. 1C is a top view of the shoe 100, FIG. 1D is a back view of the shoe 100, and FIG. 1E is bottom view of the shoe 100.

The shoe 100 may generally include a shoe upper 104 and a sole 106. The upper 104 may have a traditional shape and may be made from a combination of standard upper materials such as, for example, natural leather, synthetic leather, knits, non-woven materials, natural fabrics, and synthetic fabrics. For example, breathable, mesh, and synthetic textile fabrics made from nylons, polyesters, polyolefins, polyurethanes, rubbers, foams, and combinations thereof can be used. The material used to construct the upper may be selected based on desired properties such as breathability, durability, flexibility, comfort, and water resistance. In one example, the shoe 100 may be waterproof, and the forepart region of the upper 104 and at least an outermost layer of the upper 104 may be constructed of one or a combination of materials having water resistant properties. Additional waterproofing features (described below) may be applied in construction of the shoe 100 for providing additional waterproofing capabilities.

The forepart region of the upper 104 may include a forefoot and midfoot region that generally correspond with

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a front and a middle of the wearer's foot (e.g., the area surrounding the toes and metatarsal, phalange, and sesamoid bones and the area surrounding the arch and the navicular, cuboid, and cuneiform bones). It is understood that the heel region and forepart region (i.e., forefoot and midfoot regions) are intended to represent general areas of footwear and not demarcate precise areas. The forepart region of the upper **104** may include a vamp **108** for covering the forepart region of the foot. In some examples, the vamp **108** may cover at least a portion of a tongue member **110**. In other examples, and as shown in the various Figures, the forepart region of the upper **104** may further include an eye stay **112** that may be attached to the vamp **108** and that may cover at least a portion of the tongue member **110**.

The shoe **100** may have a medial side and a lateral side that are opposite to one another. The medial side may generally correspond with an inside area of the wearer's foot and a surface that faces toward the wearer's other foot. The lateral side may generally correspond with an outside area of the wearer's foot and a surface that faces away from the wearer's other foot. The wearer's foot may be inserted into the shoe **100** through a throat opening **114** defined in the upper **104**. In the examples, the throat opening **114** may be defined by the heel cup **102** and the tongue member **110**. In other examples, the throat opening **114** may be defined by the heel cup **102**, the vamp **108**, and the tongue member **110**. In still other examples and as illustrated in the various Figures, the throat opening **114** may be defined by the heel cup **102**, the eye stay **112**, and the tongue member **110**. A lace **118** or another tightening system may be used for tightening the shoe around the contour of the wearer's foot. The lace **118** may be threaded through eyelets or eyes of the eye stay **112**.

The heel region of the upper **104** may generally correspond with the rear portions of a wearer's foot (e.g., the area surrounding and below the Achilles tendon, the posterior of the heel, and the talus and calcaneus bones). In an example implementation, the heel region of the upper **104** includes the heel cup **102**. According to an example, the heel cup **102** may be comprised of a plurality of layers that may be molded together to form the heel cup **102**, as discussed further below. For instance, the heel cup **102** may be formed as a single molded element made up of a plurality of layers.

Example layers that may be used to form the heel cup **102** are illustrated in exploded views in FIGS. 2A and 2E-H. As should be appreciated, the example layers in FIGS. 2A and 2E-H are illustrated for better clarity and understanding of aspects of the shoe **100**. For example, the example layers in FIG. 2A are shown having shapes and contours that may not be present until after a molding process (e.g., sizes, dimensions, and configurations of the various layers, such as illustrated in FIG. 2A, may be formed through the molding process). Other example shapes and contours of layers of the heel cup **102** prior to the molding process are represented in FIGS. 2E-H. An outside (or back) view of the example layers of the heel cup **102** arranged together in a flattened form is shown in FIG. 2B, an outside (or back) view of the heel cup **102** in a curved form, for example, in the form in which the heel cup **102** may be attached to the forepart region of the upper **104**, is shown in FIG. 2C, and an inside (or front view, e.g., from the front of the shoe **100**) of the example layers of the heel cup **102** arranged together in a flattened view is shown in FIG. 2D.

With reference to FIGS. 2A-H, the heel cup **102**, and thus, the layers of the heel cup **102**, may have a medial side **50a** and a lateral side **50b** that correspond to the medial side and the lateral side of the shoe **100**. A first layer, which may be

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the outermost layer or the outer surface of the heel cup **102**, may be a non-wicking material **128**. For example, the non-wicking material **128** may be a material that does not absorb moisture as part of providing a waterproof shoe **100**. The non-wicking material **128** may wrap around the heel region and extend forward along both the medial side **50a** and the lateral side **50b**.

In some examples, a next layer of the heel cup **102** may include an outer foam layer **130**. The outer foam layer **130** may be included to provide an embossed decorative pattern in the heel cup **102**. For example, the outer foam layer **130** may be a foam pad comprised of one or a combination of pieces and may be of various shapes, sizes, and configurations. As shown in FIGS. 2B and 2C, when the various layers of the heel cup **102** are arranged and molded together, the outer foam layer **130** may provide a three-dimensional (3D) pattern on the non-wicking material **128**.

A next layer of the heel cup **102** may include a counter **132**. The counter **132** may reinforce the heel cup **102** and the upper **104** and, in some examples may limit movement of the wearer's heel. The counter **132** may wrap around the heel region and extend forward along both the medial side **50a** and the lateral side **50b**.

A next layer of the heel cup **102** may include an inner foam layer that may be molded to form one or a plurality of heel pads **134,234,334,434**. The one or more heel pads **134,234,334,434** are described in detail below with reference to FIGS. 5A-D. As shown in an outside (or back) view of an example shoe **100** worn by a wearer in FIG. 4, the heel pad(s) **134,234** may be included to add comfort around the wearer's ankle during wear. As described above, the medial malleolus formed by the lower end of the tibia is the prominence on the medial side of the wearer's ankle, and the lateral malleolus formed by the lower end of the fibula is the prominence on the lateral side of the ankle. As shown in FIG. 4, the height H_{MM} of the medial malleolus from the bottom of the wearer's foot is greater than the height H_{LM} of the lateral malleolus. In examples, the heel pad(s) **134,234,334,434** may have an asymmetrical configuration about the medial side **50a** and the lateral side **50b** of the heel cup **102** to reduce the voids of space that can normally occur around the heel region in a shoe **100** due to the asymmetrical heights H_{MM} , H_{LM} of the medial malleolus and the lateral malleolus. Accordingly, the shoe **100** may better fit the contour of the wearer's heel area for reducing foot movement during a golf swing.

Returning to FIGS. 2A-2H, a last layer, which may be the innermost layer or interior surface of the heel cup **102**, may include a lining **136**. For example, the lining **136** may be used to line the inside of the heel cup **102** and improve comfort of the shoe **100**. The lining **136** may wrap around the heel region and extend forward along both the medial side **50a** and the lateral side **50b**. As shown in FIG. 2D, when the various layers of the heel cup **102** are arranged together, the one or more heel pads **134,234,334,434** included in the inner foam layer may provide 3D asymmetrical heel padding in the lining **136**. For example, after the various layers of the heel cup **102** are molded together and assembled to the shoe **100**, the 3D asymmetrical heel padding provided by the heel pad(s) **134,234,334,434** may fit the contour of the wearer's heel area.

FIGS. 2E-H show different examples of layering of the heel cup **102** described in FIG. 2A. For example, FIG. 2E shows an example heel cup layer configuration comprising a single inner foam layer **150** that may be formed to create the one or more heel pads **134,234,334,434**. For instance, the first layer (e.g., the outermost layer or the outer surface of

the heel cup 102) may be a non-wicking material 128; the next (e.g., second) layer may include the outer foam layer 130 to provide an embossed decorative pattern in the heel cup 102 when the various layers of the heel cup 102 are arranged and molded together, the next (e.g., third) layer may include the counter 132 to reinforce the heel cup 102; the next (e.g., fourth) layer may include the single inner foam layer 150, that when molded, includes at least one or a plurality of heel pads 134,234,334,434 configured asymmetrically about the medial side 50a and the lateral side 50b of the heel cup 102 to reduce the voids of space that can normally occur around the heel region in a shoe 100 due to the asymmetrical configuration of the wearer's medial malleolus and the lateral malleolus; and the next (e.g., fifth or last) layer, which may be the innermost layer of the heel cup 102, may include the lining 136 that may be used to line the inside of the heel cup 102 and improve comfort of the shoe 100.

FIG. 2F shows an example heel cup layer configuration comprising a dual density inner foam layer 250a,b (generally, 250) that may be formed to create one or more heel pads 134,234,334,434 having dual density properties. For instance, the dual density inner foam layer 250a,b may be comprised of a medial inner foam layer 250a having a first density and a lateral inner foam layer 250b having a second density higher than the first density. For example, the higher density of the lateral inner foam layer 250b may provide additional support along the lateral side of the wearer's foot, which can reduce foot movement during a golf swing.

FIG. 2G shows an example heel cup layer configuration comprising an inner foam layer 150 and one or a plurality of heel pads 134,234,334,434 that may be pre-formed (e.g., die cut) and positioned between the inner foam layer 150 and the lining 136. For example, the one or more heel pads 134,234,334,434 may be arranged asymmetrically about the medial side 50a and the lateral side 50b of the heel cup 102, such that when the various layers of the heel cup 102 shown in FIG. 2G are molded together, the one or more heel pads 134,234,334,434 may be formed to support the wearer's foot by reduce the voids of space that can normally occur around the heel region in a shoe 100 due to the asymmetrical configuration of the wearer's medial malleolus and the lateral malleolus.

FIG. 2H shows an example heel cup layer configuration comprising a first inner foam layer 150 and a dual density inner foam layer 350a,b (generally, 350) that may be formed to create one or more heel pads 134,234,334,434 having triple density properties. For instance, the dual density inner foam layer 350a,b may be comprised of a medial inner foam layer 350a having a first density and a lateral inner foam layer 350b having a second density higher than the first density. In some examples and as shown, the medial inner foam layer 350a and the lateral inner foam layer 350b may be arranged such there a space may be defined between them. When the various layers of the heel cup 102 shown in FIG. 2H are molded together, the first inner foam layer 150 may fill the space between the medial inner foam layer 350a and the lateral inner foam layer 350b. For example, the first inner foam layer 150 may be configured to provide support in the center (i.e., between the medial side 50a and the lateral side 50b) of the heel cup 102, the medial inner foam layer 350a may be configured to form a medial side of the one or more heel pads 134,234,334,434, and the lateral inner foam layer 350b may be configured to form a lateral side of the one or more heel pads 134,234,334,434. Thus, when formed, the first inner foam layer 150 and the dual density inner foam layer 350 may provide a triple foam density heel pad

configuration. It should be understood that the above-described layers shown in FIGS. 2A-H represent only one example of layers of materials that can be used in the heel cup 102 constructions of this disclosure and other layers of materials can be used without departing from the spirit and scope of this invention.

According to an aspect, after being molded together, the heel cup 102 may be attached to the forepart region of the upper 104. In one example, and as shown in a schematic diagram illustrated in FIG. 3, the heel cup 102 may be secured to the forepart region of the upper 104 via a first stitching 140. Additionally, one or a plurality of layers of cement 144 may be applied over the first stitching 140 to fill any holes made as part of the stitching for waterproofing the seam between the heel cup 102 and the forepart region of the upper 104. In some examples, a liner 146 may be included. The liner 146 may be used to line the inside of the forepart region of the upper 104 and improve comfort of the shoe 100. For example, the liner 146 may be attached to the inner side of the forepart region of the upper 104 and extend from the toe area of the shoe 100 to the heel cup 102. The liner 146 may be secured to the heel cup 102 via a second stitching 142. The liner 146 may cover the cement 144 applied over the first stitching 140. Additionally, a piece of hot melt adhesive 148 may be applied over the second stitching 142 to cover any holes made as part of the stitching for providing a waterproof shoe 100. The hot-melt adhesive 148 may be a hot melt tape or similar material. Another layer of cement 144 may also be used to fill in the holes caused by the second stitching 142, and the adhesive 148 may be applied over the other layer of cement 144. In one example implementation, this may complete the construction of the upper 104.

It should be understood that the above-described upper 104 shown in FIGS. 1A-D, 2A-D, and 3 represents only one example of an upper that can be used in the shoe 100 constructions of this disclosure; other uppers can be used without departing from the spirit and scope of this invention.

According to an example, the sole 106 may be comprised of an outsole 116, a midsole 115, and an insole 126. Some example soles 106 that can be incorporated in the shoe 100 are described in U.S. Pat. No. 11,019,874 titled "Golf Shoe Having Outsole with All-Surface Traction Zones" filed on Dec. 20, 2018, and U.S. Patent Application No. 2020/038,3421 titled "Golf Shoe Having Midsole and Outsole for Providing Flex and Stability" filed Aug. 26, 2019, the disclosures of which are incorporated by reference herein in their entireties. The midsole 115 may be relatively lightweight and provides cushioning to the shoe 100. The midsole 115 may be made from a standard midsole material such as, for example, foamed ethylene vinyl acetate copolymer (EVA) or polyurethane. In one manufacturing process, the midsole 115 is molded on and about the outsole 116. Alternatively, the midsole 115 can be molded as a separate piece and then joined to the top surface (not shown) of the outsole 116 by stitching, adhesives, or other suitable fastening means using standard techniques known in the art. For example, the midsole 115 can be heat-pressed and bonded to the top surface of the outsole 116. It should be understood that the above-described midsole 115 shown in FIGS. 1A, 1B, 1D, and 4 represents only one example of a midsole that can be used in the shoe constructions of this invention and other midsoles can be used without departing from the spirit and scope of this invention.

In general, the outsole 116 is designed to provide stability and traction for the shoe 100. Referring back to FIG. 1E, the bottom surface of the outsole 116 is configured to contact the

ground during golf play. The golf shoe 100 may include multiple traction members 122 that protrude from the bottom surface of the outsole 116 to help provide traction between the shoe 100 and grass on the course. The traction members 122 can be of various sizes and may further be of any suitable shape including, but not limited to, rectangular, triangular, square, spherical, star, diamond, pyramid, arrow, rod, or conical-shapes. Also, the height and area of the traction members 122 may be the same or different. The bottom surface of the outsole 116 and traction members 122 can be made of any suitable material such as rubber or plastics and combinations thereof. Thermoplastics such as nylons, polyesters, polyolefins, and polyurethanes can be used. It should be understood that the above-described outsole 116 shown in FIG. 1E represents only one example of an outsole that can be used in the shoe 100 constructions of this disclosure and other outsole designs can be used without departing from the spirit and scope of this invention.

In some examples, the bottom surface of the outsole 116 may further include spikes or cleats 138 (shown in FIG. 4). The cleats 138 may provide additional traction between the shoe 100 and the ground surface. If such spikes or cleats 138 are present, they may be fastened to sockets or receptacles (not shown) in the outsole 116 or fixedly coupled to the outsole 116.

The insole 126 (shown in FIG. 1C), which may be worn inside the shoe 100, may be designed to provide cushioning or comfort for the wearer of the shoe 100. The insole 126 may be above the outsole 116 when in use. In some embodiments, the insole 126 may be designed to provide support. The insole 126 may be flexible, semi-rigid, or rigid. In some examples, the insole 126 may be removable.

The sole 106 may be attached to the upper 104 at a feather line 124 (shown in FIG. 1A), for example, using cement or the like and conventional techniques known to those skilled in the art. Prior to attachment to the sole 106, the upper 104 may be pulled onto a last, and a lasting board may be attached to the upper 104 with an adhesive. The lasting board may then be attached to the sole 106 with an adhesive for producing the shoe 100. It should be understood that the above-described sole 106 shown in FIGS. 1A-D represents only one example of a sole that can be used in the shoe 100 constructions of this disclosure; other soles can be used without departing from the spirit and scope of this invention.

With reference now to FIGS. 5A-D, different configurations of heel pads 134, 234, 334, 434 in a heel cup 102 are described. As should be appreciated, height and distance dimensions as described with respect to FIGS. 5A-D may be measured in a direction that is substantially orthogonal to a plane formed by the sole or ground when the shoe 100 is at rest. For instance, height dimensions may be measured in a substantially vertical direction. Width and/or length dimensions may be measured in a direction that is substantially parallel to the plane formed by the sole or ground when the shoe 100 is at rest. For instance, the width and/or length dimensions may be measured in a direction that is substantially orthogonal to the direction of the height dimensions. As illustrated in FIGS. 5A-D, the heel cup 102 is in a flattened form and the dimensions are shown along an oblique plane, which when the shoe 100 is assembled and at rest on or parallel to a surface, may be in a vertical position.

As shown in FIG. 5A, in one example implementation, the heel cup 102 may include a single heel pad 134, wherein the single heel pad 134 may have an asymmetrical configuration about the medial side 50a and the lateral side 50b of the heel cup 102. In some examples, the heel pad 134 may be constructed to have a medial height H_{1M} that is greater than

its lateral height H_{1L} . For instance, the medial height H_{1M} is a distance measured from a lowest point along the bottom edge of the medial side 50a of the heel pad 134 to a lowest point along the upper edge of the medial side 50a of the heel pad 134. The lateral height H_{1L} is a distance measured from a lowest point along the bottom edge of the lateral side 50b of the heel pad 134 to a lowest point along the upper edge of the lateral side 50b of the heel pad 134. In some examples, the medial side 50a of the heel pad 134 may have a height 1-6 mm or 1-3 greater than height the lateral side 50b of the heel pad 134. For instance, the medial side 50a of the heel pad 134 may have a height H_{1M} that is at least 1 mm greater than the height H_{1L} of the lateral side 50b of the heel pad 134.

The difference in height may be dependent on the size of the shoe. For instance, a larger difference in height may be present for a larger shoe, and a smaller difference in height may be present for smaller shoe. As such, the differences in height may be better represented as a ratio between the medial height H_{1M} and the lateral height H_{1L} . In some examples, ratio between the medial height H_{1M} dimension to the lateral height H_{1L} dimension is a ratio within a range of 1.05:1 to 1.5:1. As an illustrative example, for an average adult male shoe size, the medial height H_{1M} of the heel pad 134 may be between 23.5-26.5 mm, and the lateral height H_{1L} of the heel pad 134 may be 22-25 mm. Accordingly, the heel pad 134 may occupy a greater area on the medial side 50a to reduce the voids of space that can normally occur around the heel region in a shoe 100 due to the asymmetrical heights H_{MM} , H_{LM} of the medial malleolus and the lateral malleolus.

In some examples and as shown in FIG. 5A, the heel cup 102 may have a symmetrical profile, wherein the medial height H_{HCM} of a portion of the heel cup 102 that may be located below the wearer's medial malleolus (MM) may be equal to the lateral height H_{HCL} of a portion of the heel cup 102 that may be located below the wearer's lateral malleolus (LM). For instance, the medial height H_{HCM} is a distance measured from a bottom edge of the medial side 50a of the heel cup 102 to the lowest point of an upper edge of the medial side 50a of the heel cup 102. The lateral height H_{HCL} is a distance measured from a bottom edge of the lateral side 50b of the heel cup 102 to the lowest point of an upper edge of the lateral side 50b of the heel cup 102. In other examples, the heel cup 102 may have an asymmetrical profile, wherein the medial height H_{HCM} of the heel cup 102 is greater than the lateral height H_{HCL} of the heel cup 102.

In some examples, to further provide a shoe 100 that may better fit the contour of the wearer's heel area for reducing foot movement during a golf swing, the heel pad 134 may be positioned such that the medial side 50a of the heel pad 134 may be 1-6 mm or 1-3 mm higher than the lateral side 50b of the heel pad 134. For example, a first medial distance D_{1M1} , measured from the bottom edge of the medial side 50a of the heel cup 102 to the lowest point of the bottom edge of the medial side 50a of the heel pad 134, may be less than a first lateral distance D_{1L1} measured from the bottom edge of the lateral side 50b of the heel cup 102 to the lowest point of the bottom edge of the lateral side 50b of the heel pad 134. In some examples, the first medial distance D_{1M1} dimension in comparison with the first lateral distance D_{1L1} dimension is a ratio within a range of 1:1.05 to 1:1.5. As an illustrative example, for an average adult male shoe size, the first medial distance D_{1M1} of the heel pad 134 be between 21-25 mm and the first lateral distance D_{1L1} of the heel pad 134 may be between 22.5 and 26.5 mm.

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In some examples, the heel pad **134** may be further positioned such that a second medial distance D_{1M2} , measured from the lowest point along the top edge of the medial side **50a** of the heel cup **102** to the lowest point along the top edge of the medial side **50a** of the heel pad **134**, is equal to a second lateral distance D_{1L2} , measured from the lowest point along the top edge of the lateral side **50b** of the heel cup **102** to the lowest point along the top edge of the lateral side **50b** of the heel pad **134**. In one illustrative example, for an average adult male shoe size, the second medial distance D_{1M2} of the heel pad **134** and the second lateral distance D_{1L2} of the heel pad **134** may be between 5-9 mm. It should be understood that the above-described single heel pad configuration shown in FIG. 5A represents only one example of a single heel pad configuration that can be used in the shoe **100** constructions of this disclosure and other single heel pad configuration designs can be used without departing from the spirit and scope of this invention. For example, in other implementations, the heel pad **134** may be constructed such that the lateral thickness is greater than the medial thickness. In other example implementations, the heel pad **134** may be constructed to extend farther along the lateral side **50b** of the heel cup **102** than along the medial side **50a** of the heel cup **102** (e.g., between 1-3 mm). In other example implementations, the distance D_{1M2} measured from the lowest point along the top edge of the medial side **50a** of the heel cup **102** to the lowest point along the top edge of the medial side **50a** of the heel pad **134** may be less than the distance D_{1L2} measured from the lowest point along the top edge of the lateral side **50b** of the heel cup **102** to the lowest point along the top edge of the lateral side **50b** of the heel pad **134**. In some examples, the heel pad **134** may be constructed to extend farther to the top edge of the heel cup **102** on the medial side **50a** of the heel cup **102** than on the lateral side **50b** by 1-3 mm. In one illustrative example, for an average adult male shoe size, the medial distance D_{1M2} dimension in comparison with the lateral distance D_{1L2} dimension is a ratio within a range of 1:1.2 to 1:1.8. Other example asymmetrical single heel pad configurations are possible and are within the scope of the present disclosure.

In another example implementation, the heel cup **102** may include a plurality of heel pads **234a-n**, wherein the plurality of heel pads **234a-n** may have an asymmetrical configuration about the medial side **50a** and the lateral side **50b** of the heel cup **102**. As can be appreciated, utilizing a plurality of heel pads **234a-n** may allow for a reduction of material used for the inner foam layer, which can reduce the weight of the shoe **100**. For example, the plurality of heel pads **234a-n** may be positioned within the profile of the single heel pad **134**, but may be positioned with spaces between the pads. FIG. 5B shows an example heel cup **102** comprising a multiple heel pad configuration including three (3) heel pads **234a-c**. For example, a first heel pad **234a** may be positioned on the medial side **50a** of the heel cup **102**, and may be further positioned to fill a first void space defined between the heel cup **102** and the medial side of the wearer's foot below the wearer's medial malleolus. A second heel pad **234b** may be positioned on the lateral side **50b** of the heel cup **102**, and may be further positioned to fill a second void space defined between the heel cup **102** and the lateral side of the wearer's foot below the wearer's lateral malleolus. A third heel pad **234c** may be positioned in the center (i.e., between the medial side **50a** and the lateral side **50b**) of the heel cup **102**.

In some examples and as shown in FIG. 5B, the heel cup **102** may have a symmetrical profile with an asymmetrical heel pad configuration. For instance, the top edge and the

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bottom edge of the heel cup may be substantially symmetrical on the medial and lateral sides, but the heel pad configuration may be asymmetrical on medial and lateral sides. As an example, the medial height H_{HCM} of a portion of the heel cup **102** that may be located below the wearer's medial malleolus (MM) may be equal to the lateral height H_{HCL} of a portion of the heel cup **102** that may be located below the wearer's lateral malleolus (LM). For instance, the medial height H_{HCM} is a distance measured from a bottom edge of the medial side **50a** of the heel cup **102** to the lowest point of the upper edge of the medial side **50a** of the heel cup **102**; and the lateral height H_{HCL} is a distance measured from a bottom edge of the lateral side **50b** of the heel cup **102** to the lowest point of the upper edge of the lateral side **50b** of the heel cup **102**. In one illustrative example, for an average adult male shoe size, the medial height H_{HCM} of the heel cup **102** and the lateral height H_{HCL} of the heel cup **102** may be between 61.5-65.5 mm. In other examples, the heel cup **102** may have an asymmetrical profile and an asymmetrical heel pad configuration.

In some examples, the first heel pad **234a** and the second heel pad **234b** may be constructed to have the same dimensions and may be arranged such that the first heel pad **234a** is a mirror image of the second heel pad **234b** and positioned in the heel cup **102** asymmetrically to the second heel pad **234b** to reduce the voids of space that can normally occur around the heel region in a shoe **100** due to the asymmetrical heights H_{MMP} , H_{LM} of the wearer's medial malleolus and the lateral malleolus. For example, the first heel pad **234a** may be constructed to have a same medial height H_{3M} as the lateral height H_{3L} of the second heel pad **234b**. For instance, the medial height H_{3M} of the first heel pad **234a** may be measured from a lowest point along the bottom edge of the first heel pad **234a** to a lowest point along the upper edge of the first heel pad **234a**; and the lateral height H_{3L} is a distance measured from a lowest point along the bottom edge of the second heel pad **234b** to a lowest point along the upper edge of the second heel pad **234b**. In one illustrative example, for an average adult male shoe size, the medial height H_{3M} of the first heel pad **234a** and the lateral height H_{3L} of the second heel pad **234b** may be between 14.5-18.5 mm.

In some examples, the first heel pad **234a** may be further constructed to have a same medial width W_{3M} as the lateral width W_{3L} of the second heel pad **234b**. For instance, the medial width W_{3M} may be measured from the left edge (or back edge) of the first heel pad **234a** to the right edge (or front edge) of the first heel pad **234a** across a horizontal midline of the first heel pad **234a**, and the lateral width W_{3L} may be measured from the left edge (or front edge) of the second heel pad **234b** to the right edge (or back edge) of the second heel pad **234b** across a horizontal midline of the second heel pad **234b**. The widths may be measured from edge to edge following the contour of the heel cup **102**. In one illustrative example, for an average adult male shoe size, the medial width W_{3M} of the first heel pad **234a** and the lateral width W_{3L} of the second heel pad **234b** may be between 60-66 mm.

In some examples, to further provide a shoe **100** that may better fit the contour of the wearer's heel area for reducing foot movement during a golf swing, the first heel pad **234a** may be positioned such that the first heel pad **234a** may be positioned such that the first heel pad **234a** may be 1-6 mm or 1-3 mm higher than the second heel pad. For example, a first medial distance D_{3M1} measured from the bottom edge of the medial side **50a** of the heel cup **102** to the lowest point along the bottom edge of the first heel pad **234a** may be

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greater than a first lateral distance D_{3L1} measured from the bottom edge of the lateral side **50b** of the heel cup **102** to the lowest point along the bottom edge of the second heel pad **234b**. In some examples, the first medial distance D_{3M1} dimension in comparison with the first lateral distance D_{3L1} dimension is a ratio within a range of 1:1.05 to 1:1.5. As an illustrative example, the first medial distance D_{3M1} of the first heel pad **234a** may be between 36.5-40.5 mm and the first lateral distance D_{3L1} of the second heel pad **234b** may be between 34.5-38.5 mm.

In some examples, the first heel pad **234a** may be further positioned such that the first heel pad **234a** may be 1-3 mm higher than the second heel pad **234b**. For example, a second medial distance D_{3M2} measured from the lowest point along the top edge of the medial side **50a** of the heel cup **102** to the lowest point along the top edge of the first heel pad **234a** may be less than a second lateral distance D_{3L2} measured from the lowest point along the top edge of the lateral side **50b** of the heel cup **102** to the lowest point along the top edge of the second heel pad **234b**. In some examples, the second medial distance D_{3M1} dimension in comparison with the second lateral distance D_{3L1} dimension is a ratio within a range of 1:1.1 to 1:1.5. In one illustrative example, for an average adult male shoe size, the second medial distance D_{3M2} of the first heel pad **234a** may be between 5-9 mm and the second lateral distance D_{3L2} of the second heel pad **234b** may be between 7-11 mm.

It should be understood that the above-described multiple heel pad configuration shown in FIG. 5B represents only one example of a multiple heel pad configuration that can be used in the shoe **100** constructions of this disclosure and other multiple heel pad configuration designs can be used without departing from the spirit and scope of this invention. For instance, in other examples, the multiple heel pad configuration may be constructed such that the thickness of the second heel pad **234b** has a lateral thickness is greater than the medial thickness of the first heel pad **234a**. In other examples, the multiple heel pad configuration may be constructed such that the width W_{3L} of the second heel pad **234b** may be greater than the width W_{3M} of the first heel pad **234a** and may extend farther along (e.g., 1-3 mm) the lateral side **50b** of the heel cup **102** than along the medial side **50a** of the heel cup **102**. In other examples, the first heel pad **234a** may be constructed of a different material than the second heel pad **234b**. For example, the material of the second heel pad **234b** (i.e., located on the lateral side **50b** of the heel cup **102**) may have a density higher than a density of the material of the first heel pad **234a** (located on the medial side **50a** of the heel cup **102**). In other examples, other multiple heel pad configurations may be utilized.

For example and as shown in FIG. 5C, another multiple heel pad configuration may include two (2) heel pads **334a, b** having an asymmetrical configuration about the medial side **50a** and the lateral side **50b** of the heel cup **102**. According to one example, a first heel pad **334a** may be positioned on the medial side **50a** of the heel cup **102** and may be constructed to fill a first void space defined between the heel cup **102** and the medial side of the wearer's foot below the wearer's medial malleolus, and a second heel pad **334b** may be positioned on the lateral side **50b** of the heel cup **102** and constructed to fill a second void space defined between the heel cup **102** and the lateral side of the wearer's foot below the wearer's lateral malleolus. According to an aspect, the first heel pad **334a** may have a width W_{2M} greater than the width W_{2L} of the second heel pad **334b**, and may be constructed to extend around the back of the heel cup **102** (e.g., around the back of the wearer's ankle) onto a portion

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of the lateral side **50b** of the heel cup **102**. For instance, the medial width W_{2M} may be measured from the left edge of the first heel pad **334a** to the right edge of the first heel pad **334a** across the horizontal midline, and the lateral width W_{2L} may be measured from the left edge of the second heel pad **334b** to the right edge of the second heel pad **334b** across the horizontal midline. In some examples, the medial width W_{2M} dimension in comparison with the width W_{2L} dimension is a ratio within a range of 1.3:1 to 2:1.

In some examples, the first heel pad **334a** in a two (2) heel pad configuration may be positioned such that the first heel pad **334a** may be 1-3 mm higher than the second heel pad **234b**. For example, a medial distance D_{2M} measured from the lowest point along the top edge of the medial side **50a** of the heel cup **102** to the lowest point along the top edge of the first heel pad **334a** is less than the lateral distance D_{2L} measured from the lowest point along the top edge of the lateral side **50b** of the heel cup **102** to the lowest point along the top edge of the second heel pad **334b**. In some examples, the medial distance D_{2M} dimension in comparison with the lateral distance D_{2L} dimension is a ratio within a range of 1:1.05 to 1:1.5. In one illustrative example, the medial distance D_{2M} of the first heel pad **334a** may be between 5-9 mm and the lateral distance D_{2L} of the second heel pad **334b** may be between 7-11 mm.

According to another example, another multiple heel pad configuration including two (2) heel pads **434a, b** is shown in FIG. 5D. For example, a first heel pad **434a** may be positioned on the medial side **50a** of the heel cup **102** and may be constructed to fill a first void space defined between the heel cup **102** and the medial side of the wearer's foot below the wearer's medial malleolus, and a second heel pad **434b** may be positioned on the lateral side **50b** of the heel cup **102** and constructed to fill a second void space defined between the heel cup **102** and the lateral side of the wearer's foot below the wearer's lateral malleolus. According to an aspect, the first heel pad **434a** may have a width W_{2M} less than the width W_{2L} of the second heel pad **434b**, wherein the second heel pad **434b** may be constructed to extend around the back of the heel cup **102** (e.g., around the back of the wearer's ankle) onto a portion of the medial side **50a** of the heel cup **102**. For instance, the medial width W_{2M} may be measured from the left edge of the first heel pad **434a** to the right edge of the first heel pad **434a** across the horizontal midline of the first heel pad **434a**, and the lateral width W_{2L} may be measured from the left edge of the second heel pad **434b** to the right edge of the second heel pad **434b** across the horizontal midline of the second heel pad **434b**. In some examples, the medial width W_{2M} dimension in comparison with the lateral width W_{2L} dimension is a ratio within a range of 1:1.25 to 1:2.

In some examples, the first heel pad **434a** in a two (2) heel pad configuration may be positioned such that the first heel pad **434a** may be 1-3 mm higher than the second heel pad **434b**. For instance, a medial distance D_{2M} measured from the lowest point along the top edge of the medial side **50a** of the heel cup **102** to the top lowest point along the edge of the first heel pad **434a** may be less than the lateral distance D_{2L} measured from the lowest point along the top edge of the lateral side **50b** of the heel cup **102** to the lowest point along the top edge of the second heel pad **434b**. In some examples, the medial distance D_{2M} dimension in comparison with the lateral distance D_{2L} dimension is a ratio within a range of 1:1.05 to 1:1.5. In one illustrative example, the medial distance D_{2M} of the first heel pad **434a** may be between 5-9 mm and the lateral distance D_{2L} of the second heel pad **434b** may be between 7-11 mm. Other example

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asymmetrical multiple heel pad configurations are possible and are within the scope of the present disclosure.

According to an aspect, the size and configuration of the heel pad(s) **134,234,334,434** may be based on the size of the shoe **100**. In some examples, a first size and heel pad **134,234,334,434** configuration may be associated with a first range of shoe **100** sizes, a second size and heel pad **134,234,334,434** configuration may be associated with a second range of shoe **100** sizes, and so on. In other examples, heel pads **134,234,334,434** may be sized and configured based on the specific wearer. For example, the locations and sizes of the pads may be based on various measurements of a wearer's foot. One example measurement may include measurements of the distances from a standing surface to the wearer's medial malleolus and lateral malleolus or, more generally, the difference in height between the wearer's medial malleolus and lateral malleolus.

FIG. 6 depicts a flow chart having example operations of a method **600** of making a golf shoe **100** comprising an asymmetrical heel pad configuration according to an embodiment. At OPERATION **602**, a forepart region of a shoe upper **104** may be constructed. As described above, the forepart region of the upper **104** may include a vamp **108**, an eye stay **112**, and a tongue member **110**. In some examples, the vamp **108**, eye stay **112**, and tongue member **110** may be stitched or otherwise attached together.

At OPERATION **604**, a heel region of the shoe upper **104** may be constructed. According to an example, the heel region may include a heel cup **102**. Accordingly, operation **604** may include constructing the heel cup **102**. For example, in OPERATION **604**, various layers may be molded together to form the heel cup **102**, as discussed herein. One example method for constructing the heel cup **102** is described below with reference to FIG. 7.

At OPERATION **606**, the forepart region of the upper **104** and the heel cup **102** may be attached. According to one example, the forepart region of the upper **104** may be stitched or otherwise attached to the heel cup **102** to form the upper **104**.

In some implementations, at OPERATION **608**, the upper **104** may be waterproofed. For example, a method may be performed for waterproofing the stitching or attachment interface between the heel cup **102** and the forepart region of the upper **104**. One example method for waterproofing the upper **104** is described below with reference to FIG. 8.

At OPERATION **610**, the upper **104** may be lasted. According to one example, a board lasting operation may be performed, wherein the upper **104** may be stretched over a last or lasting machine, and then bonded to a paper fiber or other type of board. In some examples, a buffing process may be performed to smooth the material of the upper **104** underneath the board.

At OPERATION **612**, a sole **106** may be attached to the upper **104**. For example, the board may be bonded to an outsole **116**, and in some examples, an insole **126** may be inserted into the shoe **100**.

With reference now to FIG. 7, example operations of a method **700** of constructing an asymmetric heel cup **102** are depicted. For example, the method **700** may be performed as part of OPERATION **604** of method **600**. The method **700** may start at OPERATION **702**, where a first layer of the heel cup **102** may be arranged. For example, the first layer, which may be the outermost layer of the heel cup **102**, may be a non-wicking material **128** that does not absorb moisture as part of providing a waterproof shoe **100**.

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At OPERATION **704**, a next, or second, layer of the heel cup **102** may be arranged and placed in contact with the first layer. For example, the second layer may include an outer foam layer **130** to provide an embossed decorative pattern in the heel cup **102** when the various layers of the heel cup **102** are arranged and molded together.

At OPERATION **706**, a third layer of the heel cup **102** may be arranged and placed in contact with the second layer. For example, the third layer may include a counter **132** to reinforce the heel cup **102**.

At OPERATION **708**, a fourth layer of the heel cup **102** may be arranged and placed in contact with the third layer. For example, the fourth layer may include an inner foam layer. When molded, the inner foam layer includes at least one or a plurality of heel pads **134,234,334,434** configured asymmetrically about the medial side **50a** and the lateral side **50b** of the heel cup **102** to reduce the voids of space that can normally occur around the heel region in a shoe **100** due to the asymmetrical heights H_{MM} , H_{LM} of the wearer's medial malleolus and the lateral malleolus.

At OPERATION **710**, a fifth, or last layer, which may be the innermost layer of the heel cup **102**, may be arranged and placed in contact with the fourth layer. For example, the fifth layer may include a lining **136** that may be used to line the inside of the heel cup **102** and improve comfort of the shoe **100**. Once all the layers are arranged or placed together, a multi-layer pre-molded material is formed.

At OPERATION **721**, the various layers (e.g., the multi-layer pre-molded material) may be molded together using molding techniques to form the heel cup **102** with the asymmetric heel pads discussed herein. For instance, compression molding techniques may be used to mold the multi-layer pre-molded material into the heel cup **102**. As an example, the multi-layer pre-molded material may be placed into a heated mold cavity. The mold may then be closed and compressed to form the heel cup **102** with the heel pads positioned and formed as discussed herein.

With reference now to FIG. 8, example operations of a method **800** of waterproofing a shoe upper **104** are depicted. For example, the method **800** may be performed as part of OPERATION **608** of method **600**. The method **800** may start at OPERATION **802**, where cement **144** or another sealant may be applied over the seam or other interface (i.e., first stitching **140**) where the forepart region of the shoe upper **104** and the heel cup **102** are connected. For example, the cement **144** may be applied to the interior surface of the interface to fill any holes that may be formed in the process of attaching (e.g., stitching) the forepart region of the shoe upper **104** to the heel cup **102**. In some examples, at OPERATION **803**, heat may be applied (e.g., the upper **104** may be put into a heat tunnel) to seal the cement **144**.

At OPERATION **804**, a liner **146** may be attached to the inner side of the forepart region of the upper **104** extending from the toe area of the shoe **100** to the heel cup **102**, and then secured to the heel cup **102** via a second stitching **142**. For example, the liner **146** may cover the cement **144** applied over the first stitching **140**.

At OPERATION **805**, another layer of cement **144** may be applied to the interior surface over the second stitching **142** to fill any holes made as part of the stitching. At OPERATION **806**, heat may be applied (e.g., the upper **102** may be put into a heat tunnel) to seal the cement **144**.

At OPERATION **807**, hot melt adhesive **148** may additionally be applied over the second stitching **142** to cover the cement **144** applied over the second stitching **142**. In some examples, the hot melt adhesive may be a type of tape. At OPERATION **808**, heat may be applied to the hot melt

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adhesive to create a waterproof seal over the second stitching **142** and to provide a waterproof shoe **100**.

When numerical lower limits and numerical upper limits are set forth herein, it is contemplated that any combination of these values may be used. Other than in the operating examples, or unless otherwise expressly specified, all of the numerical ranges, amounts, values and percentages such as those for amounts of materials and others in the specification may be read as if prefaced by the word “about” even though the term “about” may not expressly appear with the value, amount or range. Accordingly, unless indicated to the contrary, the numerical parameters set forth in the specification and attached claims are approximations that may vary depending upon the desired properties sought to be obtained by the present invention.

It also should be understood the terms, “first”, “second”, “third”, “fourth”, “fifth”, “sixth”, “seventh”, “eight”, “ninth”, “tenth”, “eleventh”, “twelfth”, “top”, “bottom”, “upper”, “lower”, “upwardly”, “downwardly”, “right”, “left”, “center”, “middle”, “proximal”, “distal”, “anterior”, “posterior”, “forefoot”, “mid-foot”, and “rear-foot”, and the like are arbitrary terms used to refer to one position of an element based on one perspective and should not be construed as limiting the scope of the invention.

All patents, publications, test procedures, and other references cited herein, including priority documents, are fully incorporated by reference to the extent such disclosure is not inconsistent with this invention and for all jurisdictions in which such incorporation is permitted. It is understood that the shoe materials, designs, constructions, and structures; shoe components; and shoe assemblies and sub-assemblies described and illustrated herein represent only some embodiments of the invention. It is appreciated by those skilled in the art that various changes and additions can be made to such products and materials without departing from the spirit and scope of this invention. It is intended that all such embodiments be covered by the appended claims.

What is claimed is:

1. A golf shoe comprising:

an upper and a sole attached to the upper, the upper comprising:

a heel cup with a first surface and a second surface; and a plurality of asymmetrically shaped or positioned pads provided between the first surface and the second surface, the plurality of asymmetrically shaped or positioned pads comprising a first pad and a second pad that is entirely separated from the first pad, wherein the first pad is configured to extend along a first side of the heel cup and across a center line onto a second side of the heel cup, and wherein the second pad is positioned entirely on the second side of the heel cup between the center line of the heel cup and a front or side edge of the second side of the heel cup.

2. The golf shoe of claim 1, wherein the first pad and the second pad are arranged with a spacing defined between the first pad and the second pad.

3. The golf shoe of claim 2, wherein the spacing between the first pad and the second pad is laterally offset from the center line of the heel cup.

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4. The golf shoe of claim 2, wherein the spacing between the first pad and the second pad is positioned on the second side of the heel cup.

5. The golf shoe of claim 4, wherein the spacing is positioned entirely on the second side of the heel cup between the center line of the heel cup and the front or side edge of the second side of the heel cup.

6. The golf shoe of claim 1, wherein the first side corresponds to a medial side of the heel cup.

7. The golf shoe of claim 1, wherein the second side corresponds to a medial side of the heel cup.

8. The golf shoe of claim 1, wherein the first pad is configured to occupy a greater area on the first side than that occupied by the second pad on the second side.

9. The golf shoe of claim 1, wherein the first pad or the second pad comprises an upper edge that is positionable below a medial or lateral malleolus region of a subject's leg.

10. The golf shoe of claim 1, wherein an upper edge of the first pad is positioned at a first distance from a first lower edge portion of the heel cup on the first side of the heel cup, and wherein an upper edge of the second pad is positioned at a second distance from a second lower edge portion of the heel cup on the second side of the heel cup, wherein the first distance is different than the second distance.

11. The golf shoe of claim 1, wherein an upper edge of the first pad is positioned at a first distance from a first upper edge portion of the heel cup on the first side of the heel cup, and wherein an upper edge of the second pad is positioned at a second distance from a second upper edge portion of the heel cup on the second side of the heel cup, wherein the first distance is different than the second distance.

12. The golf shoe of claim 1, wherein a lower edge of the first pad is positioned at a first distance from a first upper edge portion of the heel cup on the first side of the heel cup, and wherein a lower edge of the second pad is positioned at a second distance from a second upper edge portion of the heel cup on the second side of the heel cup, wherein the first distance is different than the second distance.

13. The golf shoe of claim 1, wherein a lower edge of the first pad is positioned at a first distance from a first lower edge portion of the heel cup on the first side of the heel cup, and wherein a lower edge of the second pad is positioned at a second distance from a second lower edge portion of the heel cup on the second side of the heel cup, wherein the first distance is different than the second distance.

14. The golf shoe of claim 1, wherein the first pad and the second pad have different thicknesses.

15. The golf shoe of claim 1, wherein the first pad has a different length or a different width than the second pad.

16. The golf shoe of claim 1, wherein the first pad and the second pad comprise a same type of material.

17. The golf shoe of claim 1, wherein the first pad and the second pad have different densities.

18. The golf shoe of claim 1, further comprising a third pad positioned around or along at least a portion or a section of the first pad or the second pad.

19. The golf shoe of claim 18, wherein the first pad, the second pad, and the third pad form a triple density heel pad.

20. The golf shoe of claim 18, wherein the third pad is symmetrical.

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